

Adiabatic Invariant of a Harmonic Oscillator

We consider a harmonic oscillator (a mass on a spring) whose spring constant k changes slowly over time (adiabatically). The mass m is constant.

1 Adiabatic Invariant E/ω

The total energy of the oscillator is:

$$E = \frac{m\dot{x}^2}{2} + \frac{kx^2}{2}, \quad \text{where } k = k(t). \quad (1)$$

Differentiating E with respect to time:

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} + \frac{1}{2}\dot{k}x^2 = \dot{x}(m\ddot{x} + kx) + \frac{\dot{k}}{2}x^2. \quad (2)$$

The equation of motion is $m\ddot{x} + kx = 0$. Substituting this into the equation above, the first term vanishes:

$$\frac{dE}{dt} = \frac{\dot{k}}{2}x^2. \quad (3)$$

We now average this equation over one period of oscillation T . Because k changes slowly, $\dot{k}$ is approximately constant over a period. The average of the left-hand side is the time derivative of the average energy:

$$\frac{d\langle E \rangle}{dt} = \frac{\dot{k}}{2}\langle x^2 \rangle. \quad (4)$$

For a harmonic oscillator, the time-averaged potential and kinetic energies are equal:

$$\left\langle \frac{kx^2}{2} \right\rangle = \left\langle \frac{m\dot{x}^2}{2} \right\rangle = \frac{1}{2}\langle E \rangle. \quad (5)$$

From this, we can express the average of x^2 :

$$\langle x^2 \rangle = \frac{\langle E \rangle}{k}. \quad (6)$$

Substituting this result into the averaged energy equation:

$$\frac{d\langle E \rangle}{dt} = \frac{\dot{k}}{2} \cdot \frac{\langle E \rangle}{k} = \frac{\langle E \rangle}{2} \frac{\dot{k}}{k}. \quad (7)$$

This is a separable differential equation. Rearranging and integrating:

$$\frac{d\langle E \rangle}{\langle E \rangle} = \frac{1}{2} \frac{dk}{k} \Rightarrow \ln \langle E \rangle = \frac{1}{2} \ln k + \text{constant}. \quad (8)$$

Exponentiating both sides yields the adiabatic invariant:

$$\boxed{\frac{\langle E \rangle}{\sqrt{k}} = \text{constant}}. \quad (9)$$

Since the angular frequency of the oscillator is $\omega = \sqrt{k/m}$, we have $\sqrt{k} = \omega\sqrt{m}$. Therefore, the constancy of $\langle E \rangle/\sqrt{k}$ implies the constancy of a more commonly used invariant:

$$\boxed{\frac{\langle E \rangle}{\omega} = \text{constant}}. \quad (10)$$

This quantity $I = E/\omega$ is the adiabatic invariant of the system.